

# PRACTICE QUESTIONS

1. The manager sent you a sales CSV file. How would you load it into Pandas?

**Expected Topics:**

- `import pandas as pd`
  - `pd.read_csv()`
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2. Before doing any analysis, how would you quickly check whether the dataset loaded correctly?

**Expected Topics:**

- `head()`
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3. Your manager asks:

"How many rows and columns are present in this dataset?"

How would you find that?

**Expected Topics:**

- `shape`
- 

4. You received a new dataset from another team.

Before analyzing it, how would you check:

- Column names
- Data types
- Missing values

**Expected Topics:**

- `info()`
-

5. Display only the following columns from the dataset:

- CustomerName
- City
- Product

**Expected Topics:**

- Column Selection
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6. The marketing team wants to see only Laptop orders.

How would you filter them?

**Expected Topics:**

- Filtering
- 

7. Find customers who satisfy BOTH conditions:

- City = Hyderabad
- Product = Laptop

**Expected Topics:**

- Multiple Conditions
  - &
- 

8. Find customers who belong to either:

- Chennai
- Mumbai

**Expected Topics:**

- Multiple Conditions
  - |
- 

9. Create a new column called Revenue using:

Revenue = Quantity × Price

**Expected Topics:**

- Creating New Columns
- 

**10. The finance manager says:**

"Discount must be deducted from Revenue."

Create:

$\text{Revenue} = (\text{Quantity} \times \text{Price}) - \text{Discount}$

**Expected Topics:**

- Revenue Calculation
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**11. Display the products from highest price to lowest price.****Expected Topics:**

- `sort_values()`
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**12. Your manager asks:**

"Which cities are our customers ordering from?"

Return only distinct city names.

**Expected Topics:**

- `unique()`
- 

**13. The CEO asks:**

"How many unique products do we sell?"

**Expected Topics:**

- `nunique()`
-

#### 14. Which city generated the highest number of orders?

Show the frequency of all cities.

##### Expected Topics:

- `value_counts()`
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#### 15. Before preparing dashboards, identify how many missing values exist in each column.

##### Expected Topics:

- `isnull().sum()`
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#### 16. Replace missing City values with:

Unknown

and replace missing Price values using the average Price.

##### Expected Topics:

- `fillna()`
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#### 17. Remove all rows where Revenue is missing.

##### Expected Topics:

- `dropna()`
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#### 18. The same order has been recorded multiple times.

First identify duplicate records, then remove them.

##### Expected Topics:

- `duplicated()`
  - `drop_duplicates()`
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## 19. Clean the CustomerName column:

Perform all of the following:

- Convert names to uppercase
- Convert names to title case
- Find names containing the letter "a"
- Replace "Mr " with an empty string

### Expected Topics:

- String Functions
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## 20. The management team wants a summary report answering all of these questions:

- Total Revenue by City
- Average Revenue by City
- Highest Revenue by City
- Revenue by City and Product

Create the report.

### Expected Topics:

- `groupby()`
- `agg()`
- `pivot_table()`